Mini Project #3 RAG System

1. Project Context & Use Case

Goal: Identify a realistic RAG use case and clearly describe its purpose.

1.1 Problem / Context

What domain or dataset did you select?

- Domain: Classical Greek historiography, specifically Thucydides' *History of the Peloponnesian War* (Greek original + English translation + basic reference materials).

What problem or information gap does your RAG system address?

- Students and researchers constantly need to:
  - Locate specific episodes (e.g., "Corcyraean stasis," "Plague at Athens") across Books 1–8.
  - Compare Greek passages with English translations.
  - Track themes (religion, stasis, speeches, women, kairos/tuchē, etc.) across the whole work.
  - Get concise, citation-backed explanations without manually scanning commentaries.
- Current workflow: use Perseus, PDFs, and scattered lecture notes → slow, error-prone, and not context-aware.
- RAG goal: create a domain-specific research assistant that:
  - Answers questions about passages and themes in Thucydides.
  - Surfaces the relevant Greek text + English translation + section references.
  - Gives short scholarly-style summaries with proper citations to book/chapter/section.

1.2 Primary Use Case

Select one:

- Q&A over policies/manuals
- Troubleshooting / runbook lookup
- **Research assistant / document summarization**

The system primarily behaves as a research assistant and passage navigator over Thucydides.

1.3 Users / Audience

Who benefits from your RAG system?

- Primary users
  - Advanced undergraduates in Greek / Classics courses (reading Thucydides in Greek).
  - Graduate students and researchers working on topics like stasis, religion, political thought in Thucydides.
- Secondary users
  - Instructors preparing lectures, exams, and handouts.
  - Digital humanities students experimenting with RAG on classical corpora.

1.4 Success Criteria

List measurable goals (e.g., accuracy ≥ 80%, latency < 5s, citation quality).

- ≥ 85% of evaluation questions answered correctly (content + references) by human grader.
- ≥ 90% of answers include at least one correct citation (Book/Chapter/Section and/or URL).
- Average end-to-end latency ≤ 4 seconds for typical queries on small cloud instance.
- ≤ 5% hallucinated citations (references to non-existent sections).
- Users report ≥ 4/5 satisfaction in a small qualitative survey (clarity + usefulness).

2. Data & Constraints

Goal: Define the dataset, data format, and project limitations.

2.1 Corpus Details

Sources (examples / plan):

- Public-domain editions of Thucydides:
    - Greek text + English translation (e.g., Perseus Digital Library exports, public-domain Loeb-equivalent).
- Course-specific materials:
    - Instructor handouts describing themes (religion, stasis, speeches, women).
    - Short lecture notes / mini-commentaries (TXT/MD).
- Structural metadata:
    - Book / chapter / section markers (e.g., `2.47.3`).

Size (MVP target):

- Thucydides text:
    - ~8 books, ~400–500 chapter units; after chunking, 250–400 chunks.
- Commentary & course notes:
    - ~20–40 short documents (1–5 pages each).
- Total:
    - On the order of 50–100 “documents” pre-chunking → 300–600 chunks.

Formats:

- TEI-XML / XML exports (from Perseus).
- Processed to cleaned TXT / MD for content.
- Optional PDF for certain commentaries and lecture handouts.

2.2 Constraints

- local-only processing (optional toggle):
  - For demo, can run with cloud embeddings/LLM.
  - For sensitive instructor notes, provide a local-only mode (e.g., local LLM + local vector DB).
- limited budget:
  - Prefer OSS components (Chroma/FAISS, LangChain/LlamaIndex) and low-usage OpenAI plans.
- must support non-technical users:
  - Simple web UI: search bar + answer + citation panel + "show passage in Greek/English."
- must protect private data:
  - Instructors' notes and student annotations not logged to external services in local-only mode.
  - Simple authentication (course login).

3. RAG Architecture (MVP)

Goal: Outline your RAG pipeline and design decisions.

3.1 Pipeline

High-level flow:

1. Ingestion
   - Load TEI-XML or TXT versions of Thucydides.
   - Normalize into Book–Chapter–Section structure.
   - Ingest lecture notes / commentaries tagged with related sections.
2. Chunking
   - Primary unit: section-level chunks (e.g., `2.47.3`) with ~300–600 tokens.

- Use heading-based splitting (Book/Chapter/Section) + sliding window overlap.

3. Embedding
    - Encode each chunk with a multilingual or Greek-capable embedding model (e.g., OpenAI embeddings for both Greek and English).
    - Store both Greek and English text in metadata so retrieval can show aligned excerpts.
4. Vector DB
    - Store embeddings + metadata in Chroma (simple local dev) or FAISS backend.
    - Metadata: `book`, `chapter`, `section`, `language`, `source_type` (text vs notes).
5. Retrieval
    - Hybrid retrieval:
        - BM25 / keyword (English & transliterated Greek) for lexical precision.
        - Vector similarity for semantic and thematic queries ("religion and stasis in Corcyra").
6. LLM Answering
    - Retrieve top-k chunks (e.g., k=6–8).
    - Prompt LLM with:
        - Question.
        - Retrieved context (Greek + English excerpts).
        - Instructions to answer in an academic but concise style and to cite book/chapter/section.
7. Citations & UI
    - Present final answer with:
        - Footnote-style citations (e.g., "Thuc. 2.47–2.54").

- Expandable “Sources” panel showing raw text snippets and links.

3.2 Core Design Choices

- Chunking:
  - Hybrid, heading-based with token window overlap.
  - Justification: preserve natural units (sections) while overlapping to avoid splitting logical arguments or speeches.
- Embeddings:
  - Hosted embeddings (OpenAI-style multilingual) for MVP.
  - Justification: high quality, minimal ops overhead; good for both English questions and Greek/English text.
- Vector DB:
  - Chroma for development and small-scale deployment.
  - Justification: simple Python API, local file persistence, no external infra.
- Retrieval:
  - Hybrid: keyword (BM25) + dense vector, with reranking in code.
  - Justification: good for exact phrase queries (“ὀλιγαρχία” occurrences) and semantic questions (“How does Thucydides depict the breakdown of religion during stasis?”).
- LLM:
  - Hosted general-purpose LLM (e.g., GPT-5.1 class).
  - Justification: better reasoning for historical questions, low dev friction.
- Citations:
  - Required.
  - Each answer must cite at least Book.Chapter.Section and/or internal doc IDs.

4. Component Alternatives (Mini-Bakeoff)

Goal: Compare two or more tools per component and justify your choice.

Component	Option A	Option B	Criteria	Selected	Why
Vector DB	FAISS (in-memory index)	Chroma (Python + persistence)	Local, easy setup, metadata support	Chroma	Simple setup, persistent store, metadata/filtering built-in.
Embeddings	Hosted OpenAI-style embeddings	Local sentence-transformers	Quality vs. cost vs. ops complexity	Hosted	Better multilingual performance; easier for short project.
Chunking	Fixed token-size (e.g., 512 tokens)	Section-based + overlap	Alignment with text structure, recall	Section-based + overlap	Matches Book/Chapter/Section, preserves context of arguments/speeches.
Retrieval	Vector-only	Hybrid (BM25 + vector)	Exact phrase vs. semantic recall	Hybrid	Handles both “find this Greek word” and conceptual queries about themes.
LLM	Local small LLM (e.g., 7B)	Hosted GPT-class model	Accuracy, latency, resource needs	Hosted	Higher quality answers; no local GPU required.

Reranker	None	Lightweight cross-encoder	Answer quality vs. latency/complexity	None (MVP)	Simpler MVP; rely on K=8 + hybrid retrieval; add reranker later.
Citations	Simple list of doc IDs	Structured book/chapter/section	Human readability, grading ease	Structured	Matches scholarly citation; easy for instructor to verify.

5. Evaluation Plan & Results

Goal: Design and report simple test outcomes.

5.1 Test Set

- Create 18 evaluation questions spanning:
    - Factual passage queries:
        - "Where does Thucydides describe the plague at Athens?"
        - "In which sections does Thucydides discuss the Corcyraean stasis?"
    - Thematic questions:
        - "How does Thucydides connect religious breakdown to political disorder in Corcyra?"
        - "What is Thucydides' attitude toward oracles during the plague?"
    - Greek-focused queries:
        - "Explain the meaning of 'εἰκὸς' in the context of the stasis passages and locate relevant sections."
    - Cross-book / comparative:
        - "Compare Thucydides' depiction of stasis in Corcyra to how he describes civil conflict elsewhere."

- For each question, prepare a gold-standard answer:
  - Short paragraph.
  - Required citations (e.g., “Thuc. 2.47–2.54”, “3.70–3.85”) and possibly a key quote.

5.2 Metrics

- % Correct Answers:
  - Human grader labels each system answer as Correct / Partially Correct / Incorrect.
  - “Correct” = content and citations substantially match gold standard.
- % Cited Answers:
  - Answer includes at least one valid book/chapter/section or document link.
- Average Latency:
  - Measure “question submit → answer displayed” over 18 queries.

Hypothetical MVP results (first run):

- Total questions: 18
- Correct: 13
- Partially correct: 3
- Incorrect: 2
- Correct% (strict): 13/18 ≈ 72%
- Correct+Partial%: 16/18 ≈ 89%
- With citations: 17/18 (~94%)
- Average latency: 3.2 seconds

Notes:

- Errors mostly when:
    - Question is very broad ("compare all instances of stasis") and top-k misses some relevant books.
    - LLM over-generalizes and gives a plausible but not text-backed generalization.
- Citation issues:
    - 1 hallucinated section number (off by 1), flagged by grader.

6. Risks, Edge Cases & Future Work

Goal: Reflect on real-world concerns.

Edge Cases

- Large PDFs / long commentaries:
    - Scanned PDFs of commentaries may be messy; OCR errors can pollute chunks.
- Duplicate or overlapping content:
    - Same passage in multiple editions (Greek only vs Greek+English vs commentary).
    - Risk: retrieval surfaces redundant chunks instead of diverse perspectives.
- Ambiguous queries:
    - "What does Thucydides say about justice?" → extremely broad; might miss key passages.
- Greek-specific issues:
    - Inflected forms, accents, breathings, multiple spellings/transliterations.
    - Queries may be in English but the relevant passages only in Greek; need good embeddings + metadata.

Risks

- Hallucinations:
  - LLM invents interpretations not grounded in the text or cites non-existent sections.
- Outdated or partial corpus:
  - Only one translation or partial commentary; user might assume completeness.
- Overreliance by students:
  - Students may treat RAG answers as definitive scholarly interpretations instead of a starting point.
- Privacy / IP:
  - If instructor notes or not-yet-published commentary are in the corpus, must prevent them from leaking to external logs.

Future Work

- Hybrid and multi-hop retrieval:
  - Explicit multi-hop graph of themes, people, places ⇒ better answers for high-level comparative questions.
- Reranking:
  - Add a cross-encoder reranker to improve context ordering for complex queries.
- Incremental reindexing:
  - Allow instructors to upload new handouts and have them indexed in seconds without full rebuild.
- Better UI / UX:
  - Side-by-side Greek / English view for each cited passage.
  - Filters (by book, theme, language).
- Pedagogical features:

- “Show me the Greek for this answer” button.
- Mode that only surfaces passages and asks user to answer (study mode).

7. References

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